One neuron in three dimensions: demonstration notes

Page 1: Meet a neuron in 3D.

Background teaching material. Not a published research paper.

Before a neuron sends a signal, it holds a quiet electrical difference across its membrane.

In this model, that resting potential is -65 millivolts; without an input, the voltage stays there.

Rotate the schematic neuron to see its branching inputs, cell body, and myelinated axon.

Parameters throughout these notes: rest/reset = -65 mV; resistance = 10 Mohm; default tau = 20 ms. A resting neuron starts at V = -65 mV. I is applied current in nA. These are illustrative model choices, not empirical measurements. The 3D anatomy is schematic. Dendrites are drawn as branching inputs, the soma as the cell body, and six myelin sheaths around the axon. Geometry and myelin do not affect this single-compartment model.

At rest with I = 0, both the input and the voltage difference from rest are zero. There is no net change, so the model remains at -65 mV.

Membrane: The boundary that holds the voltage difference.

Dendrites / input: The branching input tree. The slider applies one lumped input current to the model, not a separate current in each branch.

Axon: The output path that carries the neuron’s signal away from its cell body.

Myelin: Insulating sheaths illustrated around the axon. Their geometry does not change this single-compartment calculation.

One neuron in three dimensions: demonstration notes
Page 2: Input arrives at the dendrites.

Background teaching material. Not a published research paper.

Input collects in the branching dendrites and flows into the cell body, the soma.

In the soma the input charges the membrane while a leak pulls it back toward rest; the time constant sets how fast it settles.

Differential equation: tau * dV/dt = -(V - Vrest) + R*I. Numerical method: forward Euler, dt = 0.1 ms. At each step, V += (-(V+65)+10*I)/tau * 0.1. Equilibrium: Vrest + R*I. There is no stochastic input. One voltage represents the whole neuron. This is not a spatial cable model. Branches and membrane pores illustrate the lesson, not individually simulated compartments or ion channels.

With R = 10 Mohm the voltage settles at Vrest + RI = -65 + 10 = -55 mV. There the leak exactly balances the input, so it stays below the -50 mV threshold.

Dendrites: The branching input tree. The slider applies one lumped input current to the model, not a separate current in each branch.

Soma: The cell body. Its single membrane voltage stands for the whole neuron in this model.

Leak: The tendency for the membrane voltage to move back toward rest.

One neuron in three dimensions: demonstration notes

Page 3: The trigger zone decides.

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Where the axon leaves the soma, the trigger zone watches the membrane voltage.

When it reaches -50 mV the model emits a spike, resets to -65 mV and holds there for 2 ms.

After the Euler voltage update, if V >= -50 mV: increment spike count; set V = -65 mV; set refractory timer = 2 ms. While that timer is positive, decrement it by dt (clamped at zero) and hold V = -65 mV. A +20 mV graphical marker records an event, not an action-potential waveform. Each spike triggers a 60 ms illustrative transit animation along the 3D axon. This duration is a visualization choice, not a computed or measured conduction speed.

At 1.4 nA the voltage settles at -65 + 14 = -51 mV and never reaches -50 mV. In this model the input must exceed 1.5 nA to fire.

Trigger zone: The axon hillock, drawn where the axon begins. The model fires when the voltage reaches -50 mV.

Spike: A discrete output event: a pulse leaves the trigger zone and travels along the axon.

Reset: After a spike, voltage returns to -65 mV before integration resumes.

PAPER-TO-WORLD / AUTHORED DEMONSTRATION NOTES

One neuron in three dimensions: demonstration notes
Page 4: Follow the signal down the axon.

Background teaching material. Not a published research paper.

Stronger input shortens the climb to threshold, so pulses leave the trigger zone more often.

Our simulation produced 33 spikes at 2 nA and 63 at 3 nA in the first 1 s (0?1000 ms); the axon and terminal tags count the same window.

Reference targets measured over 1000 ms at tau=20 ms: I=0 nA -> 0 spikes; I=1 nA -> 0 spikes; I=2 nA -> 33 spikes; I=3 nA -> 63 spikes. Approximate continuous-model rate for I>1.5 nA: f=1000/(tau*ln(R*I/(R*I-15))+2) Hz. Finite observation time and discrete Euler updates cause small differences from this formula. Passing numerical agreement is not validation against a published paper or biological data. The 3D presentation uses exactly the same discrete update rule as the Python and 2D references.

This single-compartment model has one membrane voltage and no cable, so myelin and branch length cannot change the count. Our simulation gives 33 spikes at 2 nA and 63 at 3 nA in the first 1 s (0?1000 ms), matching the checked Python reference.

Axon terminals: Where pulses arrive. The tag shows the firing rate from the first 1 s.

Interspike interval: Time between two spikes; shorter intervals mean a faster rhythm.

Myelin: Insulating sheaths illustrated around the axon. Their geometry does not change this single-compartment calculation.